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Kelas : TPL A2

Praktikum 11

A. Node

```
Pertemuan 11_Vincensius Alfa S_J0403251075.py > ...
1 # =====
2 # Vincensius Alfa Setyawan
3 # J0403251075
4 # TPL A2
5 # =====
6
7 # =====
8 # MENENTUKAN NODE (VERTEX)
9 # =====
10 nodes = [
11     "Router1",
12     "Switch1",
13     "Switch2",
14     "PC1",
15     "PC2",
16     "Server"
17 ]
18
```

PROBLEMS 11 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Kuliah\Semester 2\Algoritma Stryktur Data\Praktikum\Praktikum 11> & C:/Users/Adi/Python.exe "d:/Kuliah/Semester 2/Algoritma Stryktur Data/Praktikum/Praktikum 11/Pertemuan 11_Vincensius Alfa S_J0403251075.py"
=== NODE ===
Router1
Switch1
Switch2
PC1
PC2
Server
```

B. Edge

```
20 # =====
21 # MENENTUKAN EDGE
22 # =====
23 edges = [
24     ("Router1", "Switch1"),
25     ("Router1", "Switch2"),
26     ("Switch1", "PC1"),
27     ("Switch1", "PC2"),
28     ("Switch2", "Server"),
29     ("Switch2", "PC2")
30 ]
31
```

PROBLEMS 11 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Kuliah\Semester 2\Algoritma Stryktur Data\Praktikum\Praktikum 11> & C:/Users/Adi/Python.exe "d:/Kuliah/Semester 2/Algoritma Stryktur Data/Praktikum/Praktikum 11/Pertemuan 11_Vincensius Alfa S_J0403251075.py"
=== EDGE ===
('Router1', 'Switch1')
('Router1', 'Switch2')
('Switch1', 'PC1')
('Switch1', 'PC2')
('Switch2', 'Server')
('Switch2', 'PC2')
```

C. Adjacency List

```
33 # =====
34 # ADJACENCY LIST
35 # =====
36 graph = {
37     "Router1": ["Switch1", "Switch2"],
38     "Switch1": ["Router1", "PC1", "PC2"],
39     "Switch2": ["Router1", "Server", "PC2"],
40     "PC1": ["Switch1"],
41     "PC2": ["Switch1", "Switch2"],
42     "Server": ["Switch2"]
43 }
44
45 print("=== NODE ===")
46 for node in nodes:
47     print(node)
48
49 print("\n=== EDGE ===")
50 for edge in edges:
51     print(edge)
52
53 print("\n=== ADJACENCY LIST ===")
54 for node, connections in graph.items():
55     print(node, "->", connections)
56
```

PROBLEMS 11 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\Kulyahh\Semester 2\Algoritma Stryktur Data\Praktikum\Praktikum 11> & C:/Users/Admin/OneDrive/Desktop/Python/Python 3.9.0/Python 3.9.0\python.exe "d:/Kulyahh/Semester 2/Algoritma Stryktur Data/Praktikum/Praktikum 11/Pertemuan 11/Adjacency List/Adjacency List.py"

=== ADJACENCY LIST ===

Router1 -> ['Switch1', 'Switch2']
Switch1 -> ['Router1', 'PC1', 'PC2']
Switch2 -> ['Router1', 'Server', 'PC2']
PC1 -> ['Switch1']
PC2 -> ['Switch1', 'Switch2']
Server -> ['Switch2']

D. Adjacency Matrix

```
58 # =====
59 # ADJACENCY MATRIX
60 # =====
61 matrix = []
62
63 for node1 in nodes:
64     row = []
65
66     for node2 in nodes:
67         if node2 in graph[node1]:
68             row.append(1)
69         else:
70             row.append(0)
71
72     matrix.append(row)
73
74 print("\n=== ADJACENCY MATRIX ===")
75 print("      ", " ".join(nodes))
76
77 for i in range(len(matrix)):
78     print(nodes[i], matrix[i])
79
```

PROBLEMS 11 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\Kulyahh\Semester 2\Algoritma Stryktur Data\Praktikum\Praktikum 11> & C:/Users/Admin/OneDrive/Desktop/Python/Python 3.9.0/Python 3.9.0\python.exe "d:/Kulyahh/Semester 2/Algoritma Stryktur Data/Praktikum/Praktikum 11/Pertemuan 11/Adjacency List/Adjacency Matrix.py"

=== ADJACENCY MATRIX ===

Router1 Switch1 Switch2 PC1 PC2 Server

Router1 [0, 1, 1, 0, 0, 0]
Switch1 [1, 0, 0, 1, 1, 0]
Switch2 [1, 0, 0, 0, 1, 1]
PC1 [0, 1, 0, 0, 0, 0]
PC2 [0, 1, 1, 0, 0, 0]
Server [0, 0, 1, 0, 0, 0]

E. Hubungan Node

```
81 # =====
82 # HUBUNGAN ANTAR NODE
83 # =====
84 print("\n=== HUBUNGAN ANTAR NODE ===")
85
86 for node, connections in graph.items():
87     print(node, "terhubung dengan", ", ".join(connections))
88
```

PROBLEMS 10 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\Kulyahh\Semester 2\Algoritma Stryktur Data\Praktikum\Praktikum 11> & C:/Users/Admin/AppData/Local/Programs/Python/Python39-64/Python.exe "d:/Kulyahh/Semester 2/Algoritma Stryktur Data/Praktikum/Praktikum 11/Pertemuan 11_Vincensius Alif" === HUBUNGAN ANTAR NODE === Router1 terhubung dengan Switch1, Switch2 Switch1 terhubung dengan Router1, PC1, PC2 Switch2 terhubung dengan Router1, Server, PC2 PC1 terhubung dengan Switch1 PC2 terhubung dengan Switch1, Switch2 Server terhubung dengan Switch2

F. Analisis

```
90 # =====
91 # ANALISIS SINGKAT
92 # =====
93 print("=== ANALISIS ===")
94
95 print("1. Jenis Graph : Undirected Graph")
96 print("2. Alasan memilih edge : Edge merepresentasikan koneksi fisik antar perangkat")
97
98 print("3. Representasi yang lebih cocok : Adjacency List")
99 print("    Karena lebih hemat memori")
100
101 print("4. Kategori Graph : Sparse Graph")
102 print("    Karena jumlah edge tidak terlalu banyak")
103
104
```

PROBLEMS 11 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\Kulyahh\Semester 2\Algoritma Stryktur Data\Praktikum\Praktikum 11> & C:/Users/Admin/AppData/Local/Programs/Python/Python39-64/Python.exe "d:/Kulyahh/Semester 2/Algoritma Stryktur Data/Praktikum/Praktikum 11/Pertemuan 11_Vincensius Alif" === ANALISIS === 1. Jenis Graph : Undirected Graph 2. Alasan memilih edge : Edge merepresentasikan koneksi fisik antar perangkat 3. Representasi yang lebih cocok : Adjacency List Karena lebih hemat memori 4. Kategori Graph : Sparse Graph Karena jumlah edge tidak terlalu banyak

G. Kesimpulan

```
105 # =====
106 # KESIMPULAN
107 # =====
108 print("=== KESIMPULAN ===")
109 print("Graph dapat digunakan untuk memodelkan hubungan antar perangkat jaringan komputer.")
110 print("Adjacency list lebih efisien untuk graph ini.")
```

PROBLEMS 11 OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

PS D:\Kulyahh\Semester 2\Algoritma Stryktur Data\Praktikum\Praktikum 11> & C:/Users/Admin/AppData/Local/Programs/Python/Python39-64/Python.exe "d:/Kulyahh/Semester 2/Algoritma Stryktur Data/Praktikum/Praktikum 11/Pertemuan 11_Vincensius Alif" === KESIMPULAN === Graph dapat digunakan untuk memodelkan hubungan antar perangkat jaringan komputer. Adjacency list lebih efisien untuk graph ini.

H. Gambar Graph

```
# =====  
# BENTUK GRAPH  
#  
#           Router1  
#          /      \  
#         /        \  
#      Switch1 ---- Switch2  
#         /      \  
#      PC1  PC2-----/  Server  
#  
# =====  
#
```